

Analysis of Current Methods of Rice Crop Spraying in India for Smallholder Farmers

GDBP Market Study

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Introduction

Smallholder farmers, farmers that own less than 2 hectares of land[1], represent 85.2% of total farms worldwide, with the largest concentration in India and China[2]. The main crop type cultivated in these farms are grains with the majority being rice.

Rice is the edible seed of the grass species: Oryza Sativa and is primarily farmed in flooded paddies. The seeds are sown in prepared beds, after which they are transplanted onto a rice paddy when they reach 25 to 50 days of age. The field is then flooded with 5 to 10 cm of water and the rice is left to grow until it is harvested after roughly 105 to 180 days later[3]. Spraying the crop and the paddies with fertilizers, herbicides, fungicides and insecticides is an integral part of rice farming as it facilitates crop growth, prevents plant infection and repels pests or insects.

The business requirement is to create a drone system capable of spraying rice crops at a pricepoint that is accessible to smallholder farmers in India. As drone technology has yet to enter this market, the main competitors will be the traditional methods of spraying crops: manual spraying and tractor spraying.

This market study will provide an overview of the current spraying methods used in India and how they, and the market, may evolve in the future.

Manual Spraying

Manual spraying is the cheapest but slowest method of spraying crops, and is the most common method used by smallholders in India[4]. It consists of a farmer, or hired worker (1), manually spraying crops using a sprayer (2) connected to a tank (3), usually worn as a backpack, that contains the product that is being sprayed (4). This equipment is commonly known as a knapsack, capacities range from 5 to 25 litres of product and can weigh up to 20kg[6].



Figure 1: Indian farmer spraying crops manually[5]

Covering rate, Cost and Efficiency

As the farmer is required to walk around the entire field to spray the corps and periodically return to refill the tank with product, manual spraying is considered a labour-intensive process. This leads to an average covering rate of 0.4 to 0.8 hectares per day per worker[6]. For farms that are less than one hectare of land, the farmer often performs the spraying. For larger farms, working hands, usually consisting of other people from the same village, are hired at a cost of ₹162 to ₹405 per acre, with the equipment supplied by the farmer[4].

Hand spraying offers flexibility with upfront investment as knapsacks have a wide range of price points, as seen in **Table 1**.

Sprayer Type	Price Range (INR)	Common Use Case
Manual (Hand-Operated)	₹800 – ₹2,500	Small plots & marginal farmers.
Battery-Operated	₹1,800 – ₹4,500	Small to medium paddies
Petrol (2-Stroke)	₹6,500 – ₹10,000	Large scale fields requiring high pressure and speed.
Petrol (4-Stroke)	₹8,500 – ₹18,000+	Professional use; more fuel-efficient and durable.

Table 1: Table showing price ranges of different knapsack types[7]

The tank size of knapsack sprayers does not scale with cost, with 16 and 18 litres being the most common tank size across all tiers of knapsacks. The area one tank covers depends on how much product is needed per hectare, on average it ranges from 0.03 to 0.05 hectares for a 16-litre tank[4]. Smallholder farmers typically own manually-pumped or battery-operated knapsacks as they are more cost effective. Petrol powered knapsacks are typically owned by farmers with more land.

Manual spraying however can result in less efficient application of product on crops, leading to increased spraying product costs. A study found that a UAV spraying 30L of product per hectare achieved the same level of weed control as manual spraying with 500L of product used per hectare[8]. This is a result of over-spraying, which can be a common occurrence, especially when farmers are working in extreme heat for long hours.

Health Risks

Manual spraying of pesticides and herbicides requires PPE (Personal Protection Equipment) to protect the worker from the chemicals that are being sprayed. The IRRI (International Rice Research Institute) recommends clothing that covers the entire body (ideally a purpose-made suit), gloves, boots and a respirator to avoid ingesting the spray droplets while spraying the crops[9]. PPE costs can range from ₹300 for incomplete or basic kits to over ₹1,500 for professional kits[10]. Unfortunately, few Indian farmers use the correct PPE when working with pesticides, a study showed that 25% of Indian farmers (out of a sample of 170) do not use PPE and only 58% cover their nose and mouth, most often with a cloth[11]. Another study (results shown in **Figure 2**) found that 63% of farmers suffer from one or more symptoms directly after working with herbicides and pesticides with more severe symptoms being associated with a lack of PPE[12]. These chemicals also harm the surrounding community as they are not disposed of correctly or stored in unsuitable ways[11, 12]. The conclusion was that PPE must be more widely adopted between farmers to mitigate the negative impacts on themselves and their communities.

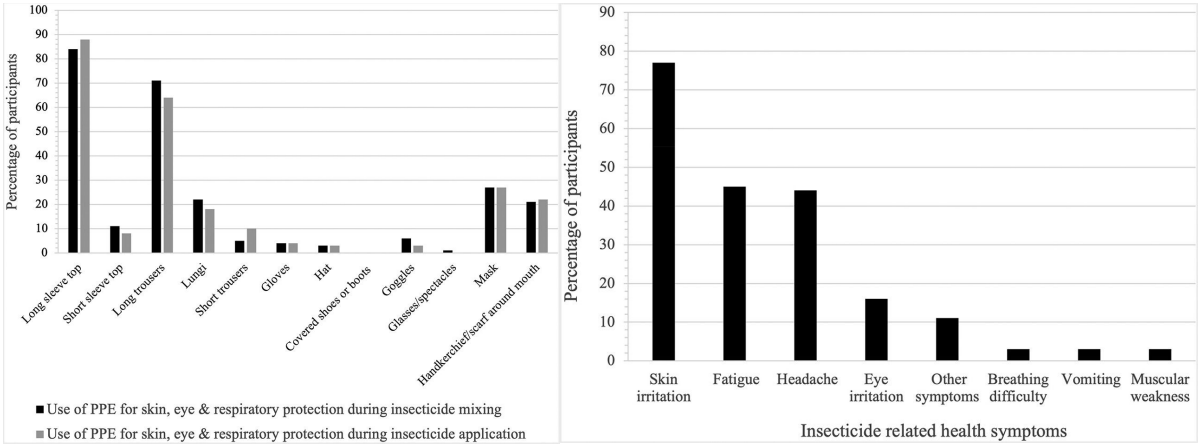


Figure 2: Figures 1 and 2 from "Insecticide exposure and associated acute health effects in farmers in a rice growing district of India" showing PPE used by farmers and number of participants experiencing symptoms related to pesticide use[12]

Despite manual spraying presents health risks, a lower covering rate, higher workload and is inefficient with the product used per area, it is still the preferred method of spraying for smallholder farmers as it requires little upfront investment and training.

Tractor / Boom Spraying

The second most common spraying method is tractor spraying (spraying equipment (1) mounted on a tractor (2)) as it offers a more efficient way to spray crops while still being at an accessible price point. However, studies found that only 20% to 30% of smallholder farmers own agricultural machinery and the vast majority uses manual spraying[13].



Figure 3: Indian farmer using a tractor to spray rice crops[14]

Covering rate, Cost and Efficiency

Many factors influence the covering rate of tractor sprayers, mainly: the width of the boom, the flowrate and the speed of the tractor. A study evaluating a low-horse-power tractor-operated sprayer in groundnut fields found that the effective covering rate was 0.90 ha/hr at a slow speed of 1.5 km/h, and had a maximum of 2.06 ha/hr when driven at 3.5 km/h giving it a maximum spreading area of 20 ha for a 10 hour day (accounting for refills)[15].

Similarly to knapsack sprayers, tractor spraying equipment has a wide price range depending on their size and tank capacity as seen in **Table 2**.

Equipment Type	Tank Capacity	Price Range (INR)	Common Use Case
Tractor Mounted Power Sprayer	500L – 1000L	₹35,000 – ₹75,000	General crops; high-pressure but manually guided
Basic Boom Sprayer	200L – 400L	₹65,000 – ₹1,58,000	Small field crops; automatic horizontal coverage
Commercial Boom Sprayer	600L – 1100L	₹1,95,000 – ₹3,50,000	Large-scale wheat, soy, or cotton farming
Self-Propelled Boom Sprayer	1000L	₹20,000 (second hand) – ₹18,00,000	High-value vineyards and fruit orchards

Table 2: Table showing price ranges of different tractor spraying systems[16]

Data comparing tractor spraying efficiency (whether fields are over or under-sprayed) to other spraying methods was not found. However, it can be assumed that fields will be sprayed more uniformly if tractor speed and spraying pressure are kept constant throughout the spraying operation, potentially leading to a higher product use efficiency.

Disadvantages of using a tractor to spray

The major disadvantage of tractor spraying is the fact that the machine will have to be driven over the field, or that special tracks will have to be made for it to drive on, sacrificing farmable land. Driving over land will compact and damage the soil, impacting fertility and destroying the habitat of beneficial organisms[17]. Furthermore, heavy machinery may get stuck in the flooded rice paddies and would destroy the levelled surface needed for equal water distribution. Tractors also require room to manoeuvre which is sometimes not available with smallholder rice farms.

Sustainability and welfare risks are also present as tractors use conventional fuels to operate, which produce greenhouse gasses. Tractor spraying offers less chemical exposure compared to manual spraying, however PPE is still recommended (both for spraying and for preparing the chemicals) but is being used scarcely[12]. The use of a tractor may reduce the harmful effects associated with chemical exposure, but will not eliminate them completely.

Tractor spraying is used for many other crop types but has yet to find popularity among rice farmers. Even if the equipment prices decrease as more second-hand equipment becomes available, the inherent downsides associated with tractor spraying will still keep smallholder rice farmers from purchasing and using tractors to spray their crops.

How the Situation may Evolve

The unique selling points (USPs) for the business case are the increased welfare of the farmer, by reducing the working hours but mainly reducing exposure to the harmful chemicals, and increased efficiency, both in time taken to spray a given area and more effective use of pesticides thereby reducing long-term costs. The current situation benefits the chosen USPs, however, some USPs may lose their appeal as time progresses.

Bio foods and Biopesticides / Bioherbicides

The emergence of demand for Bio foods, foods that are grown without synthetic fertilizers, pesticides or genetically modified organisms, has seen an increased amount of organic farming[18]. Although organic farming will produce lower crop yields, more smallholders are beginning to farm in a sustainable and eco-friendly way because their crop can then be sold at a higher price[19]. This will have an impact on the demand for crop spraying. which will see a decrease, but it will not eliminate it completely. Most smallholder farmers that begin farming organically are part of large cooperatives that support their farming, independent farmers are more likely to continue with traditional farming practices as they are less risky and produce higher yields.

Another way to make farming more sustainable is to use bio-pesticides and herbicides. These chemicals are less harmful to the environment and less toxic to the farmers as they are composed of naturally occurring substances that are recognized as safe for human contact[20]. PPE will still be needed but the discomfort to farmers using bio-pesticides will be less compared to farmers using traditional chemicals. This could have an impact on the farmer welfare USP as health hazards will be reduced. However, biopesticides target specific animal species, therefore multiple products will have to be sprayed to protect the crops from multiple pests[21]. This may increase the demand for an automated spraying system, as more spraying operations will be needed, but the increase in pesticide cost may limit a farmer's possibility to buy or rent equipment.

Climate Change

Climate and fatigue may have a large impact on manual spraying, rice is grown between March and September, meaning that crops are sprayed during the hottest period of the year. The primary regions where rice is grown in India has seen between 16 and 6 heat waves in the summer of 2023[22]. Farmers avoid labour intensive work around the peak hours of the day to avoid heat stroke. With raising global temperatures, the times of day where farmers can do labour-intensive work may decrease and working at night is not always possible due to visibility limitations. This could either lead to a manual spraying

rate decrease or to farmers working under extreme heat to the detriment of their health. This would provide prime opportunity for the drone spraying business, specifically those advertising farmer welfare. However, climate change may also bring water shortages leading to farmers having to rent water pumps to flood their fields, thereby limiting the extra monetary resources they have[23]. Extreme heat and extreme weather due to climate change may also have an impact on rice production, making smallholder rice production more difficult and risky, leading to farmers being reluctant in investing in equipment or a service[23].

Urbanisation and Government Subsidies

Urbanisation may have both a positive and negative impact on smallholder farmers. Currently around 42% of India's workforce is in agriculture[24], but with the increase in urbanisation, rising from 32% in 2013 to 36.36% in 2023[25], the number of agricultural workers is decreasing. A continued migration to urban areas, and decrease in the agricultural workforce, may have multiple impacts. Labour costs may increase as there are less people that can be hired for work, this is assuming that demand remains unaltered. The workforce may decrease in quality in the long-term as the farmers get old and there are fewer younger farmers available to replace them as the younger generation has moved to the cities. This would decrease agricultural output and may lead to government intervention, in the form of subsidies, if the export demands are not met as India's main exports are agricultural goods[19]. Government programs and external financial services are already in place to support farmers, easing purchasing of equipment and agricultural products, however these may increase as time goes on[26,23].

These factors would be extremely beneficial for drone spraying businesses as automation increases output. Increased access to technology has popularized drones in the agriculture business and this trend may rise if more companies get government support. Furthermore, as the younger generation moves to the cities and begins working in service and industry jobs, a large quantity of them may send money back to their families in rural areas, giving these farmers an extra source of income that could be invested in agricultural equipment. However, competition will increase as other businesses will seize this business opportunity.

Conclusion

The current methods used by smallholders for crop spraying prioritise low cost over every other aspect of the operation and thus far have been adequate for use on small farms. However, there is a business opportunity as certain demands are not being met: efficiency, application rate, sustainability and farmer welfare. However, price point will be the driving factor in determining whether a new product or service will be successful.

More business opportunities may arise as the Indian population moves to urban areas, transforming the agricultural economic landscape, and as farming becomes more eco-friendly and more integrated with technology. However, it would also lead to a more competitive market where a business' USPs become more important.

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